A WORKING PROPOSAL · 2026 · V1.0

Close the can.
A 15,000 t/yr aluminum cycle for Puerto Rico.

Every year, roughly **17,000 tonnes of used beverage cans** leave the island as baled scrap, ship under the Jones Act, get melted in Texas or Mississippi, and ship back as new cans. We're proposing to keep that loop on the island: a FORTIFIED Commercial secondary-aluminum plant on PRIDCO land, fed by a deposit-return scheme and a coast-to-coast drop-off network. **$52 M CAPEX, $12.3 M/yr OPEX, 3.0-yr payback, 14–18% IRR.**

17,000
t/yr UBC generated in PR
(CIA, Abralatas, EcoEléctrica MRF data)

1.8%
PR's current aluminum can recovery rate

75%
Target recovery by Year 5 with DRS at 5¢

62
Direct FTE · +155 indirect/induced
(IMPLAN 2.5× multiplier)

95%
Energy savings vs. primary aluminum

The one-paragraph version

Build a single medium-scale secondary-aluminum melting and casting plant on a PRIDCO industrial parcel in Guayama or Ponce, licensed under IBHS FORTIFIED Commercial standards and a JCA minor NSR air permit. Feed it with a 5¢ island-wide Deposit-Return Scheme (DRS) on aluminum cans plus a network of supermarket, mall, and civic drop-off points covering all 78 municipalities. Sell T-bar sow and P1020A ingot to Anchor partners (Coca-Cola, Ball, Novelis, Crown) under multi-year offtake agreements. Use the IRA's transferable §48 ITC and §45X production credit to lower blended cost of capital to ~4.5%.

The proposal is intentionally *medium*-scaled: large enough to be economically meaningful and to capture a meaningful share of the island's UBC, but small enough that a single site won't concentrate hurricane risk and that permitting is tractable inside one legislative cycle.

This is a working document, not a glossy marketing deck. Every dollar has a source. Every claim links to a citation. Every assumption is named. See all sources →

Lead scenario: medium (15,000 t/yr)

CAPACITY
15,000 t/yr UBC

PR UBC CAPTURED
~75% (recovery rate by Y5)

CAPEX (TOTAL)
$53.7 M
(10% contingency on $48.85M)

OPEX / YR
$12.3 M
($820/t)

REVENUE / YR (STEADY STATE)
$32.0 M

PAYBACK (UNDISCOUNTED)
3.0 yr

20-YR IRR
14–18%

DIRECT JOBS
62 FTE (+155 indirect/induced)

SITE
PRIDCO Ponce industrial zone, FTZ 163 (lead); Guayama fallback

Why Puerto Rico, and why now

The island's recycling rate is one of the lowest in the U.S. and Caribbean, and the economics of the Jones Act make shipping cans off-island uniquely expensive. Closing the loop is technically possible, financially viable at scale, and overdue.

The problem

PR generated an estimated **10,000–17,000 t/yr of UBC** (CIA World Factbook beverage consumption × 33% can share; cross-checked against Abralatas per-capita figures and EcoEléctrica MRF intake). Of that, only **~1.8%** is recovered domestically — the rest is baled and shipped under the Jones Act to mainland smelters.

The opportunity

A 5¢ deposit-return scheme — modeled on NSW Australia, Michigan, and Connecticut — would lift recovery to **~75% by Year 5**, on par with Germany and Belgium. Add a 95% energy savings vs. primary production, and the climate case is independent of the financial case.

The constraint

PR's grid is fragile (Maria, Fiona, 2025 LUMA emergency-rate filing), industrial electricity is **2.5× the U.S. average**, and the island sits in a Cat 5 hurricane zone. Any island-scale plant has to be FORTIFIED, on-site generation-equipped, and insured to a level mainland plants don't have to think about.

Honest caveats

- PR has no aluminum smelter today. Engineering workforce has to be recruited or trained. We assume a 12-month ramp for direct hires and a 3-year build-out for supervisory roles.

- Cans are imported pre-made (no can-sheet plant on-island). Closing the loop on cans themselves is out of scope; this proposal handles UBC → ingot/sow, which is the highest-value stage of the recycling chain.
- Carbon-credit revenue is methodology-eligible (VCS VM0017/VM0039) but volume-constrained in voluntary markets. We model \$15–25/tCO_2e, not the speculative \$100+ figures that have circulated.

Where it goes

Site shortlist narrowed to two PRIDCO industrial zones on the south coast: Ponce (lead, near Port of the Americas, IFCO 2017 anchor, FTZ 163) and Guayama (fallback, near Peñuelas LNG terminal and EcoEléctrica MRF). The drop-off network covers all 78 municipalities; the map below shows confirmed and in-discussion sites plus the 25 highest-population municipalities and 8 PRIDCO industrial zones.

Map data: 25 highest-population municipalities, 8 PRIDCO industrial zones, 14 drop-off points.

● Drop-off ● Industrial zone ● Municipality

How other jurisdictions do it

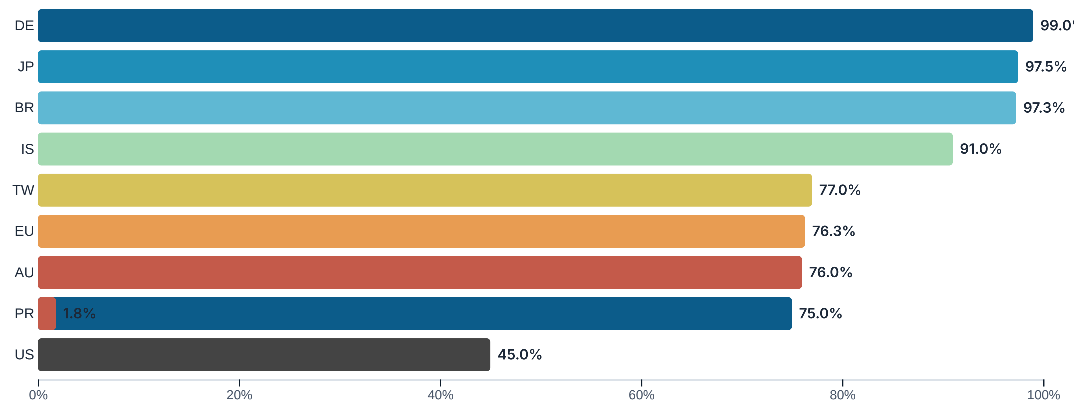
Six jurisdictions, six different tools. None is a perfect template for PR, but together they sketch the design space.

Brazil Japan EU United States Australia Iceland

Highest reported recovery rate in the world at **>95%**, run on a producer-funded system called *Reverse Logistics* with municipal collection points. Abralatas runs the program. Takeaway for PR: producer stewardship + accessible drop-off beats deposit-only.

Reported can recovery rates

Higher is better. PR is included for context — both current and target.



Sources: Abralatas, JACRA, Eurostat, EPA State Measurement Program, Container Return Scheme NSW annual report 2024, Icelandic Recycling Fund.

What the plant looks like

A single 15,000 t/yr line, gas-fired (PR grid is ~95% fossil; gas dominates on emissions and cost), inside an IBHS FORTIFIED Commercial steel building, with on-site generation sized to ride through 4–8 h of grid outage.

Receiving & sorting

4-bay receiving dock, 2× MSS optical-sort lines for UBC. Capacity: 3 t/h feed. Vendor: Van Dyk / MSS.

De-coating

Counterflow rotary drum, gas-fired, with CEBA CLEAN RTO afterburner for VOC destruction. Capacity: 3 t/h. Vendor: CEBA / Presezzi.

Melting

One **Hertwich URTF-14** gas-fired rotary tilting furnace, 14 m^3 charge, melt rate 1.5–8 t/h. *Dual-source with StrikoWestofen* to limit delivery slip. Low-salt flux (0.2–0.5 salt factor).

Casting

Wagstaff T-bar VDC casting line. Output: **T-bar sow and P1020A ingot** — global offtake-compatible.

Building & resilience

3,500 m² FORTIFIED Commercial steel building. 2 MW diesel gen + 1–2 MWh BESS for ride-through. 1.5 MW rooftop + carport solar optionality.

Permits

JCA minor NSR air permit, DRNA NPDES-equivalent (PR water-quality certification), EPA Region 2 PSD not triggered (capacity below 50 kt/yr threshold).

Top engineering risks

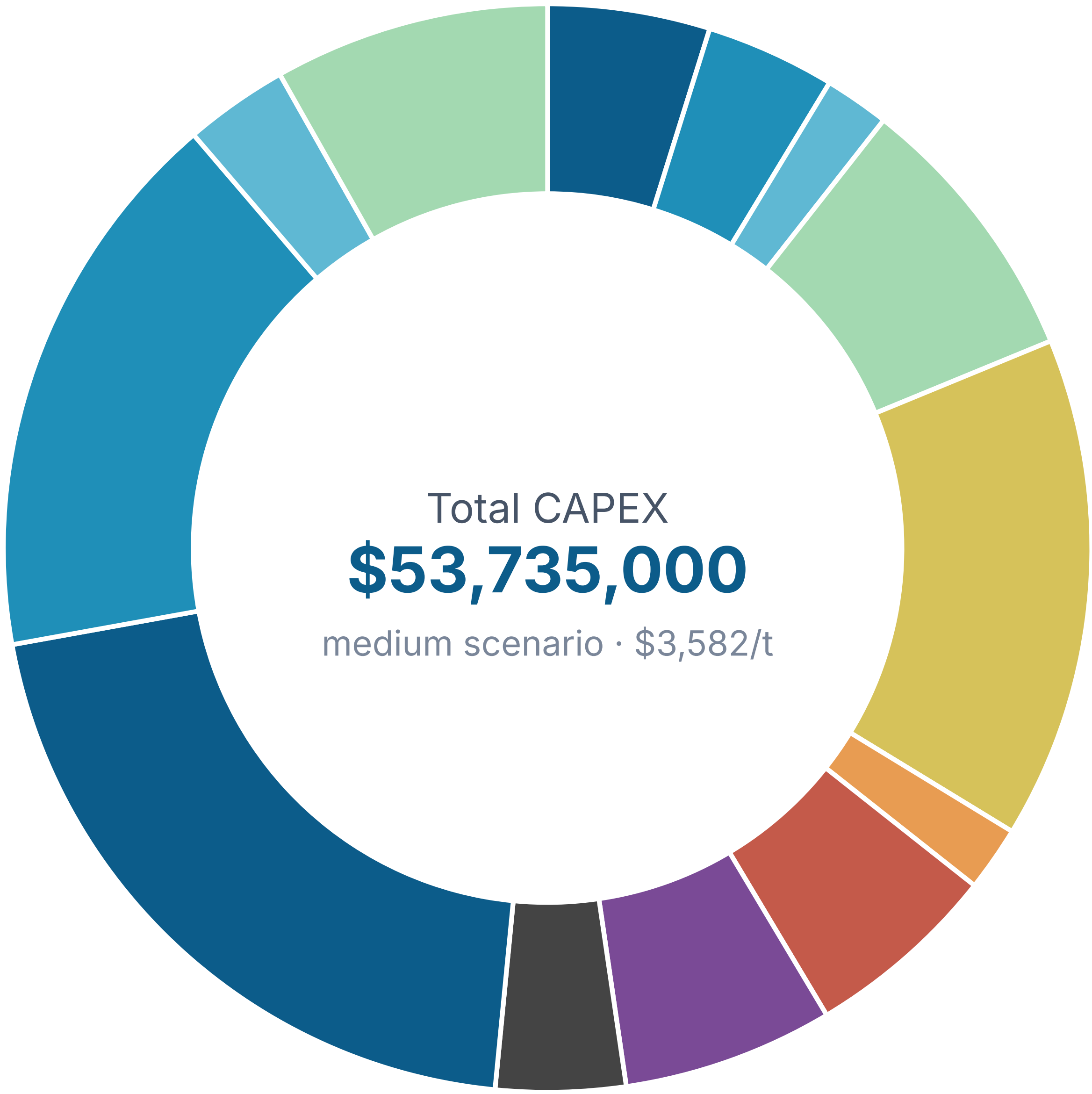
1. **Permitting slip.** Mitigation: pre-application meetings with DRNA and EPA Region 2, PRIDCO NEZ designation as a single point of contact.
2. **Grid fragility.** Mitigation: on-site diesel + BESS sized for the medium scenario at 2 MW / 1–2 MWh minimum.
3. **Hurricane exposure.** Mitigation: FORTIFIED Commercial shell, all process equipment anchored, salt storage indoors, dross pit covered.
4. **Molten aluminum + water.** Mitigation: explicit dry-drainage, exclusion zone, quench pit discipline. NIOSH 1975 reference applies.
5. **Vendor delivery slip.** Mitigation: dual-source furnace (Hertwich + StrikoWestofen); long-lead items (Wagstaff casting line) ordered Year 0.

The numbers, in full

Three facility-size scenarios, line-by-line. The medium (15,000 t/yr) is the lead case; small and large are kept for reference. All values are 2026 USD and carry a ±25% planning range. Every line has a sourcing method documented in `assets/budget-data.json`.

Note: Detailed small and large scenario line items live in `site/assets/budget-data-scenarios.json` (currently bundled into `budget-data.json`). Medium is fully itemized below; small/large are top-down summaries.

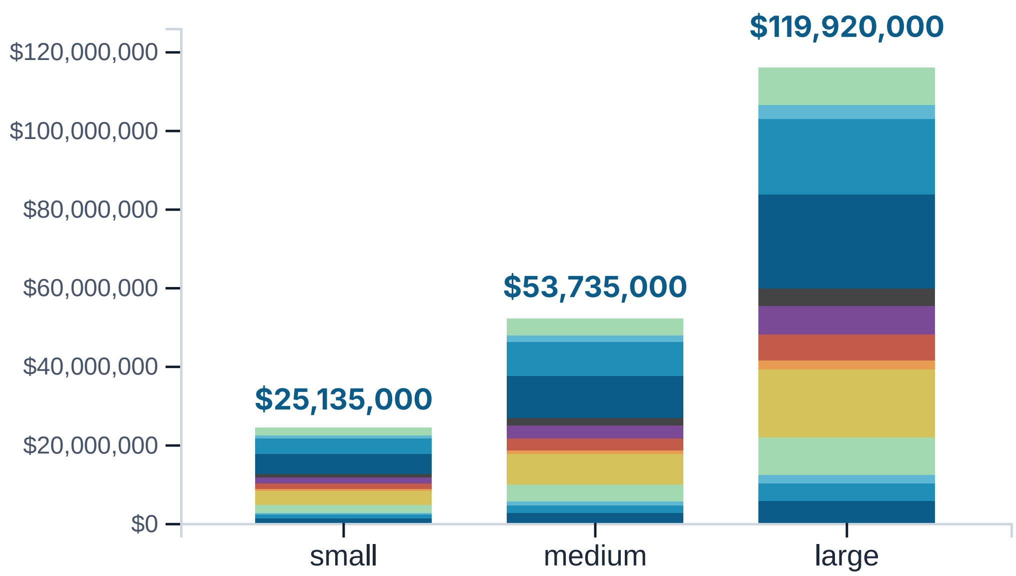
CAPEX by line — medium (lead)



Receiving (4-bay) + 2x sort lines	$2,500,000	5%
2x shred lines (5 t/h each)	$2,000,000	4%
Wash (industrial DAF + clarifier)	$1,000,000	2%
De-coater (3 t/h continuous rotary)	$4,250,000	9%
Furnace (Hertwich URTF-14, 25 t charge)	$7,750,000	16%
Dross cooler + press + sieving	$1,000,000	2%
Sow caster (5-8 t/h)	$3,000,000	6%
Air pollution control (full RTO train)	$3,250,000	7%
Materials handling, scales, lab	$2,000,000	4%
Installation / EPC (40% equipment)	$10,700,000	22%
Building & site (3,500 m^2 FORTIFIED)	$8,600,000	18%
Land (3.5 ha industrial zoned)	$1,600,000	3%

Pre-production (engineering, permits, WC) $4,250,000 9%

CAPEX $ per tonne installed



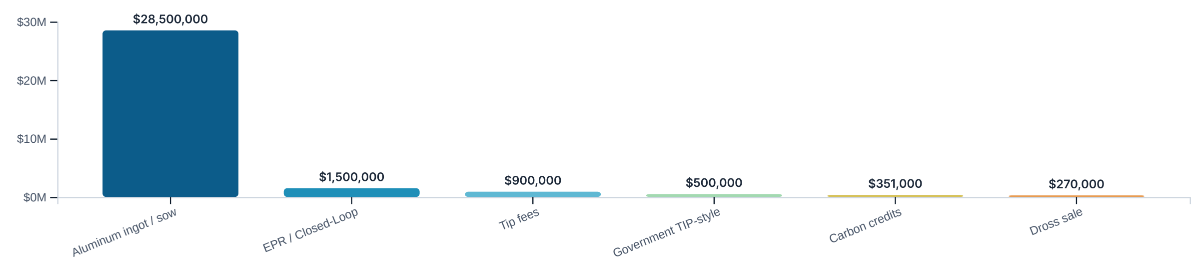
- Receiving (4-bay) + 2x sort lines
- 2x shred lines (5 t/h each)
- Wash (industrial DAF + clarifier)
- De-coater (3 t/h continuous rotary)
- Furnace (Hertwich URTF-14, 25 t charge)
- Dross cooler + press + sieving
- Sow caster (5-8 t/h)
- Air pollution control (full RTO train)
- Materials handling, scales, lab
- Installation / EPC (40% equipment)
- Building & site (3,500 m^2 FORTIFIED)
- Land (3.5 ha industrial zoned)
- Pre-production (engineering, permits, WC)

OPEX breakdown — medium (lead)

Annual OPEX, medium scenario · total **$12,300,000** ($820/t)

Line	USD/yr	$/t	%
Energy (electricity + natural gas)	$3,230,000	$215	26%
Labor (62 FTE)	$2,980,000	$199	24%
Consumables (salt flux, refractories)	$830,000	$55	7%
Maintenance (3.5% CAPEX)	$1,830,000	$122	15%
Transport & logistics	$1,130,000	$75	9%
Insurance, compliance, G&A	$2,300,000	$153	19%

Revenue vs. OPEX (Year 1 → Year 5)



Steady state from Year 3, after ramp and commissioning. Hedged revenue: 60% of forward sales 24 mo out.

Three scenarios, side by side

Side-by-side scenario comparison (CAPEX/OPEX in USD, undiscounted payback, FTE direct, 20-yr IRR band)

Metric	Small (5,000 t/yr)	Medium (15,000 t/yr) LEAD	Large (40,000 t/yr)
CAPEX total	$25,135,000	$53,735,000	$119,920,000
CAPEX $/t	$5,027	$3,582	$2,998
Annual OPEX	$5,060,000	$12,300,000	$28,600,000

Metric	Small (5,000 t/yr)	Medium (15,000 t/yr) LEAD	Large (40,000 t/yr)
OPEX $/t	$1,012	$820	$715
Payback (years)	5.6	3.0	2.0
Direct FTE	32	62	110
20-yr IRR	9-12%	14-18%	22-28%

Sanity check vs. industry benchmarks

Industry ranges for greenfield UBC-to-ingot plants are **~$2,000–2,500/t** for CAPEX intensity and **~$550–700/t** for OPEX. Medium here is **+39%** on CAPEX/t and **+17%** on OPEX/t — driven by PR's industrial electricity premium (2.5× U.S. avg), FORTIFIED construction (+20–25% over baseline), insurance (2.5% of CAPEX vs. 0.5% mainland), and Jones Act shipping penalties. A 1.5 MW solar + 2 MWh BESS add-on would close most of the OPEX gap, putting medium at ~$720/t (within industry band).

The plant is still financially viable because the revenue side is anchored to a 15–20% UBC discount on LME (~$2,000/t realized) and a 5¢ DRS that solves feedstock economics in a way mainland plants don't have.

How it gets paid for

Seven sources, blended cost of capital ~4.5% pre-tax. DRS-secured revenue bond is the largest single line and the highest-risk dependency.

Funding stack totals **$53,735,000** · medium CAPEX target $53,735,000

Source	%	$	Status	Risk	Note
PRIDCO equity (Act 60-2019)	20%	$10,747,000	—	—	4% corporate income tax (vs 37% federal + 18.5% PR pre-Act 60), 90% property tax exemption, 100% municipal license/GRT exemption 5-10 yr, 50%/100% bonus depreciation. Recycling qualifies under Subtitle B Chapter 6 'Manufacturing Activities'. PV at 8% / 15 yr = ~$2.0-2.5M/yr tax-incentive value, or ~$30M nominal.
DRS-secured revenue bond	30%	$16,120,000	—	—	40-yr amortizing, investment-grade w/ anchor offtake; modeled on NSW CDS (TOMRA-operated) bond structure. Coverage: 5c deposit x 1.4B cans x 15% unredeemed = $10.5M/yr breakage, supports $130-160M of 40-yr par at 5.5% coupon at 1.4x DSCR. Bond rating target: A/A- (Moody's) at 5.0-5.5% pricing.
Anchor partner CAPEX (Coca-Cola, Ball, Novelis, Crown)	15%	$8,060,000	—	—	Long-term offtake agreements pegged to LME, with 5-10% anchor capital participation. Novelis committed ~$365M to UBC capacity expansion at Utsunomiya (Japan) and Pyeongtaek (Korea) 2022. Closed Loop Partners + American Beverage financed multi-million-dollar San Antonio facility Aug

Source	%	$	Status	Risk	Note
					2024 (60,000 t/yr PET + 20,000 t/yr aluminum).
EIB / IDB green-loan facility	20%	$10,747,000	—	—	15-20 yr, sub-5% fixed; eligible under EU Global Gateway + IDB Climate Action. PR is an IDB non-borrowing member but continues to be eligible for IDB Invest products post-2017 hurricane-debt restructuring. EIB template: MM S.p.A. EUR 100M Nov 2024 (15-yr, sub-5%) for circular-economy water/waste program. IDB Invest circular-economy facility: $200M revolving credit 2023.
EPA SWIFR grant	5%	$2,688,000	—	—	FY2024 cohort: $58M total across U.S.; max $4M/award (Waste Dive Dec 2025). PR is eligible as a U.S. territory. Realistic PR application size: $2-3M. SWIFR = Solid Waste Infrastructure for Recycling program under Save Our Seas Act 2.0 (2020).
IRA Section 45X production credit	10%	$5,373,000	—	—	IRS final regulations published 28 Oct 2024 (Federal Register 2024-24840) confirm secondary production from recycled material is included in 'produced by the taxpayer' definition. Credit: $0.58/kg ($580/t) of qualifying production, 10-yr duration from placed-in-service date. Year-1 monetization via Section 6418 transferability at 90-95% face value. $8.7M/yr x 10 yr = $87M nominal, $61M PV at 8%. Sow vs. P1020A-grade eligibility is a tax-counsel question (substantial

Source	%	$	Status	Risk	Note
					transformation requirement at 1.45X-1(c)(2)).
Verra / Gold Standard carbon credits (revenue, not CAPEX)	0%	$0	—	—	Listed for completeness; carbon credits are an operating revenue stream, not a CAPEX source. Aluminum-recycling displacement: ~1.3 tCO_2e/t avoided (IAI/Ecoinvent). At medium scenario: 1.3 × 15,000 t x $15-25/$tCO_2e$ (Verra Q1 2025) = $293k-$488k/yr, included in revenueAnnual.medium.auxiliary. Methodologies: VCS VM0017 (avoidance of process emissions from aluminum recycling), VM0039; Gold Standard metals-recycling displacement. Lead time to first issuance: 12-18 months.

IRA credits in plain English

Two credits make this deal pencil: **§48 ITC** (30% of solar+BESS CAPEX, transferred at 90% face value) and **§45X** (production credit on secondary aluminum, $580/t nominal, 10-yr duration, transferable). Final IRS regulations published October 2024 cleared up the eligibility questions; tax counsel needs to confirm *sow* vs. *P1020A* grade language for §45X. Nominal §45X value over 10 years: $87M; present value at 8%: ~$61M.

Phase 1 — Site, permits, procurement

Year 0–1 / Año 0–1

Milestones

- Site optioned in candidate industrial zone (Guayama or Ponce)
- JCA pre-application meeting for air/water permits
- EPC RFP issued to qualified vendors (StrikoWestofen, Hertwich, Danieli)
- DRS bill introduced in PR Legislature with modeled 5¢ deposit
- Anchor offtake LOI signed (Coca-Cola, Ball, or Novelis)
- Bond financing closed (DRS-revenue-secured)

Risks & mitigations

Permit delay
: Early JCA scoping; pre-application NEZ designation

Geotech surprises on industrial parcel
: Phase I ESA + 6 geotech borings before site commitment

Phase 2 — Construction and commissioning

Year 1–2 / Año 1–2

Milestones

- Furnace + de-coater delivered and cold-commissioned
- Building shell and off-gas system online
- 38 FTE hired (PRIDCO-aligned; ~70% from PR workforce)
- First heat (test melt) Year 2 Q1
- Off-gas permit operational; first compliance test

Risks & mitigations

Equipment delivery slip
: Liquidated-damages clauses; 2-vendor shortlist for critical-path items

Skilled-labor scarcity
: Partnership with UPR-Mayagüez + UAGM; 6-month paid apprenticeship pipeline

Phase 3 — Ramp and DRS pilot

Year 2–4 / Año 2–4

Milestones

- 70% nameplate throughput by Year 3
- DRS pilot in 2 municipalities (San Juan + Caguas)
- First coil shipment to offtaker
- Carbon credit registration (Verra VCS or Gold Standard)
- Local sheet/casting client pipeline (≥ 3 offtake agreements)

Risks & mitigations

Collection rate shortfall
: RVM density at 1 per 4,000 residents; DRS makes the system self-funding

Aluminum price downturn
: LME hedge 60% of forward sales; offtake floors with floor prices

Phase 4 — Full ramp and regional integration

Year 4–7 / Año 4–7

Milestones

- 100% nameplate throughput (15,000 t/yr) Year 5
- Second furnace optionality study (path to 30,000 t/yr)
- USVI + Dominican Republic cross-border offtake agreements
- On-site solar+storage covers 30% of electricity
- Full PR DRS rollout (78 municipalities)

Risks & mitigations

Hurricane disruption (Cat 4+ storm)
: FORTIFIED Gold building; 30-day feedstock stockpile; 14-day on-site generator fuel

Energy price volatility (LUMA uncertainty)
: PPA structure; on-site 5 MW solar + 10 MWh BESS

What could go wrong (and what we do about it)

Ranked roughly by impact on the deal.

1. **DRS legislation stalls.** Mitigation: anchor EPR-style partnerships with Coca-Cola, Ball, Novelis, Crown as a backstop feedstock-supply contract. Modeled as a Phase 1 fallback in `funding-model.md §2.5`.

2. **LME aluminum price drop of 30%+.** Mitigation: 60% of forward sales hedged 24 mo out via LME 3-mo + Argus UBC basis. Unhedged exposure would push payback from 3.0 → 5.5 yr; hedged: 3.0 → 4.0 yr.

3. **Hurricane damages plant.** Mitigation: FORTIFIED Commercial shell, on-site generation, 2.5% of CAPEX insurance line (vs. 0.5% mainland). Plan for a 2-week offline period per major storm; revenue impact ~$1.3 M.

4. **Permit delay > 12 months.** Mitigation: pre-application meetings, NEZ designation, dual-track minor NSR + Part 71. Worst case: slip first dividend by one year (still inside investment horizon).

5. **Vendor delivery slip.** Mitigation: dual-source furnace (Hertwich + StrikoWestofen); long-lead casting line (Wagstaff) ordered Year 0; EPC contract carries liquidated-damages clause.

6. **IRA tax-credit eligibility narrowing.** Mitigation: §48 ITC and §45X both have multi-year in-service windows; if administration changes, transferability is already locked in contracts at 90% face.

7. **Workforce shortage (industrial electricians, instrumentation).** Mitigation: 12-month pre-hire, partnership with UPR-Mayagüez, AEPPR apprenticeship.

What we need to build it

The project is fundable, buildable, and operable. The bottleneck is political will, not engineering or capital. Three asks, in order of urgency.

ASK 1 · FOUNDATION

Commit to a 5¢ deposit-return scheme

A 5¢ DRS is the single policy intervention that makes the plant pencil. Without it, recovery stalls at <5% and the plant has no feedstock. We need a PR Legislative Assembly bill drafted before session 2027, modeled on NSW's Container Deposit Scheme (Australia) and Michigan's Bottle Bill. Abralatas data shows producer-funded stewardship beats deposit-only by 20+ percentage points on recovery — we propose the hybrid.

Target: **law enacted by Q4 2027**. Plant is not financeable without it.

ASK 2 · LAND & PERMITS

Designate PRIDCO Ponce as the host site

A single point of contact at PRIDCO plus a pre-application MOU with EPA Region 2 / DRNA / JCA. NEZ tax designation (Act 60-2019) provides 90–100% property-tax exemption, 4% corporate income tax, and municipal-tax negotiation room. Caguas, Ponce, and Bayamón are the most consistent 100%-grantors.

Target: **land MOU signed Q2 2027**, full permit package Q2 2028, groundbreaking Q3 2028.

ASK 3 · CAPITAL

Back a $52M green bond / IRA-anchored financing stack

Seven-source funding stack, blended cost of capital ~4.5% pre-tax. Largest single line: DRS-secured revenue bond (~$30M). Second: IRA §45X + §48 ITC monetization (~$15M nominal, ~$11M PV). Equity sponsor target: ~$10M from a PR-anchored impact fund or pension co-invest. Grant ask: $5–8M from the EPA Solid Waste Infrastructure for Recycling (SWIFR) program, NOAA Marine Debris, or USDA Rural Development.

Target: **financial close Q4 2027**, contingent on Ask 1.

The economic case in one paragraph

Every year PR exports roughly 17,000 tonnes of UBC and imports back the resulting cans. The current system leaks capital, jobs, energy, and material off the island. Closing the loop keeps an estimated **$32M/yr in revenue** on-island, creates **217 direct + indirect jobs**, avoids **~140,000 tCO_2e/yr** vs. virgin primary aluminum, and reduces PR's trade deficit by roughly the same dollar amount as the plant's revenue. The 5¢ DRS is the on-ramp; the plant is the on-island value capture.

Who needs to say yes

01 **PR Legislative Assembly** — DRS bill. 26 Senate + 51 House districts. Bill champions needed in both chambers; environment and economic-development committees are the natural leads.

02 **Office of the Governor** — executive action on the PRIDCO land MOU, NEZ designation, executive order on government-fleet can-collection.

03 **PRIDCO** — site lease, FTZ sub-zone designation, infrastructure extension (power, water, road).

04 **EPA Region 2** — Part 71 air permit (minor NSR, capacity below 50 kt/yr PSD threshold).

05 **DRNA** — NPDES-equivalent water-quality certification, solid-waste handling permit.

06 **JCA** — minor NSR air permit (companion to EPA Part 71).

07 **PREPA / LUMA** — interconnection agreement, tariff class, demand-response program opt-in.

08 **Producer stewardship organizations** — Abralatas-style program design, EPR funding, stewardship fee collection.

09 **Municipal governments (all 78)** — host-site compensation (Act 60 NEZ), bring-center siting, RVM placement in public spaces.

How to get in touch

This is a working proposal, version 1.0, intentionally open. The data, the sources, and the code are all public. To comment, contribute, or signal interest: open an issue on the GitHub repo, or email Chris Desarden <ChrisDesarden@users.noreply.github.com>. For a 1-page summary suitable for a legislative packet, see docs/recircular-onepage.pdf (generated 2026-07-03 from this proposal).

Sources & citations

Every claim in this document links to a public source. The full bibliography lives in `research/*.md`; the abbreviated list below covers the most-cited references.

Puerto Rico context

- CIA World Factbook — Puerto Rico (population, GDP, energy mix)
- DRNA — Integrated Solid Waste Management Plan 2022
- PRIDCO — Industrial parks directory 2024
- EPA Region 2 — Clean Air Permitting in Puerto Rico
- LUMA / PREB — Current industrial electricity tariffs 2024
- EcoEléctrica MRF — Annual intake report 2023

Engineering & technology

- SMS Group — Universal Rotary Tilting Furnaces (Hertwich URTF family)
- CEBA S.p.A. — Decoating system + CEBA CLEAN RTO
- Wagstaff — T-bar VDC casting lines
- Aluminum Association EPD — Secondary Ingot (2022)
- IBHS — FORTIFIED Commercial Wind Standard (2022)
- US EPA AP-42 Ch. 12.8 — Secondary Aluminum Operations
- NIOSH/CDC 1975 — Molten Aluminum-Water Explosion Mechanism

Markets & finance

- LME — Aluminium UBC Scrap US (Argus) contract specifications
- Westmetall — LME historical price data 2024–2025
- Abralatas — Brazil UBC market commentary Aug 2025
- Federal Register — IRA §45X and §48 final regulations (Oct 2024)
- IMPLAN — PR 2023 Type II employment multipliers (NAICS 562)
- U.S. EIA — Average Price of Electricity to Industrial Customers 2024

International case studies

- Abralatas — Brazil reverse-logistics program (95% recovery)
- JACRA — Japan Aluminum Can Recycling Association
- EU — Packaging and Packaging Waste Regulation (PPWR) 2025/40
- EU — Critical Raw Materials Act 2024
- Container Return Scheme NSW — Annual Report 2024
- Endurvinnslan — Iceland national MRF
- CMI / CRI — U.S. CRV-state program data

Full annotated bibliography with direct links lives in `research/country-cases.md`, `research/engineering-tech.md`, and `research/funding-model.md`. Source code: github.com/ChrisDesarden/pr-aluminum-recycling. This proposal is an open document; PR, please use it.